Azure Site Recovery – Cross-Region VM Replication and Test Failover

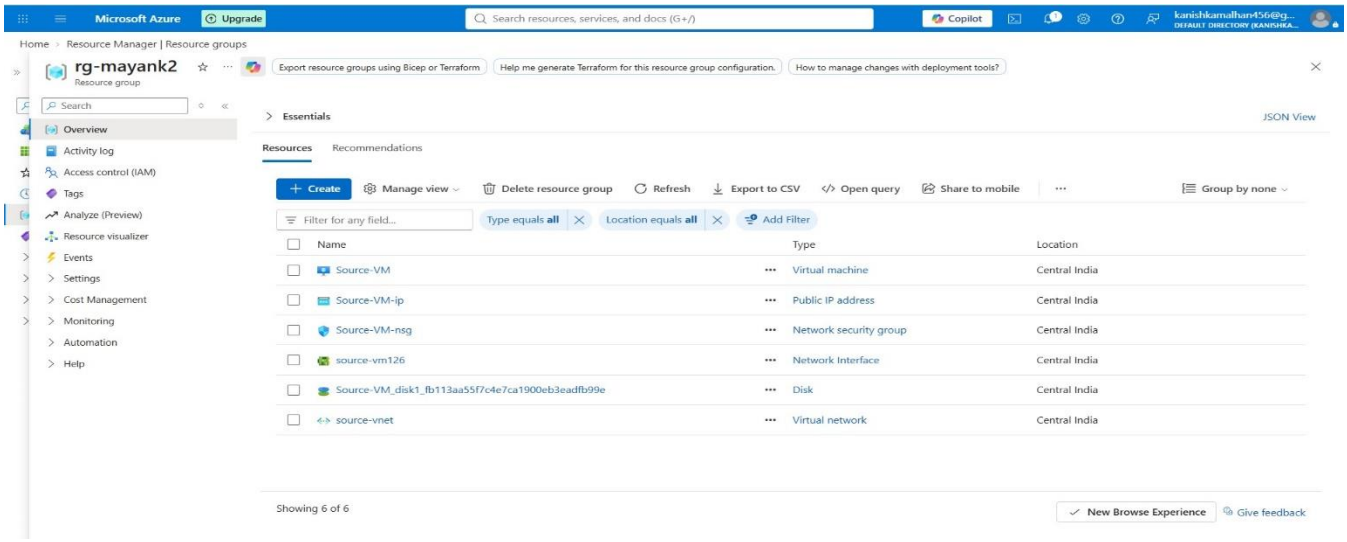
Objective

To configure Azure Site Recovery for a virtual machine in Central India, replicate it to South India, perform a test failover, verify the test virtual machine, and clean up the test environment.

Procedure

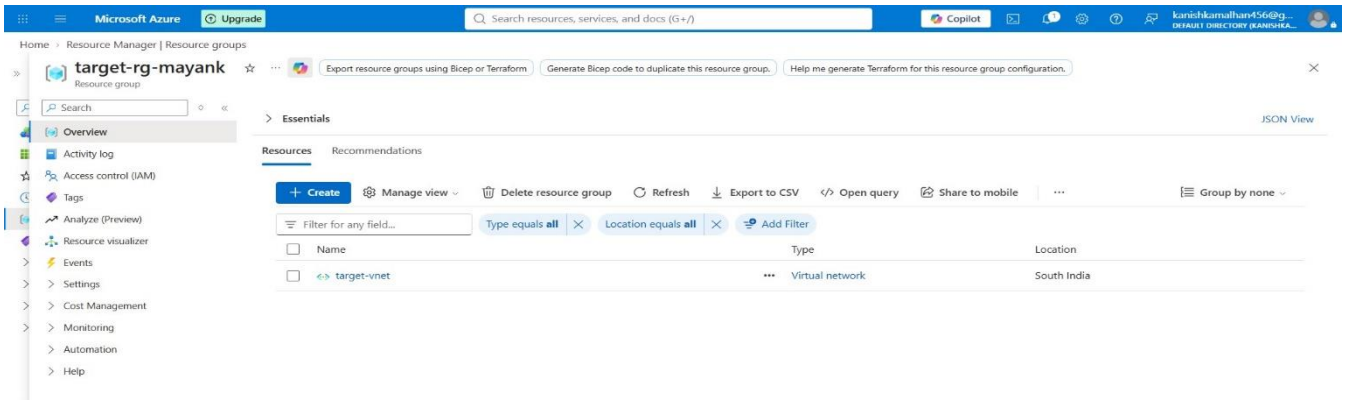
Step 1: Review the Source Resource Group

Open the source resource group, rg-mayank2, and verify that Source-VM and its associated network and disk resources are available in Central India.



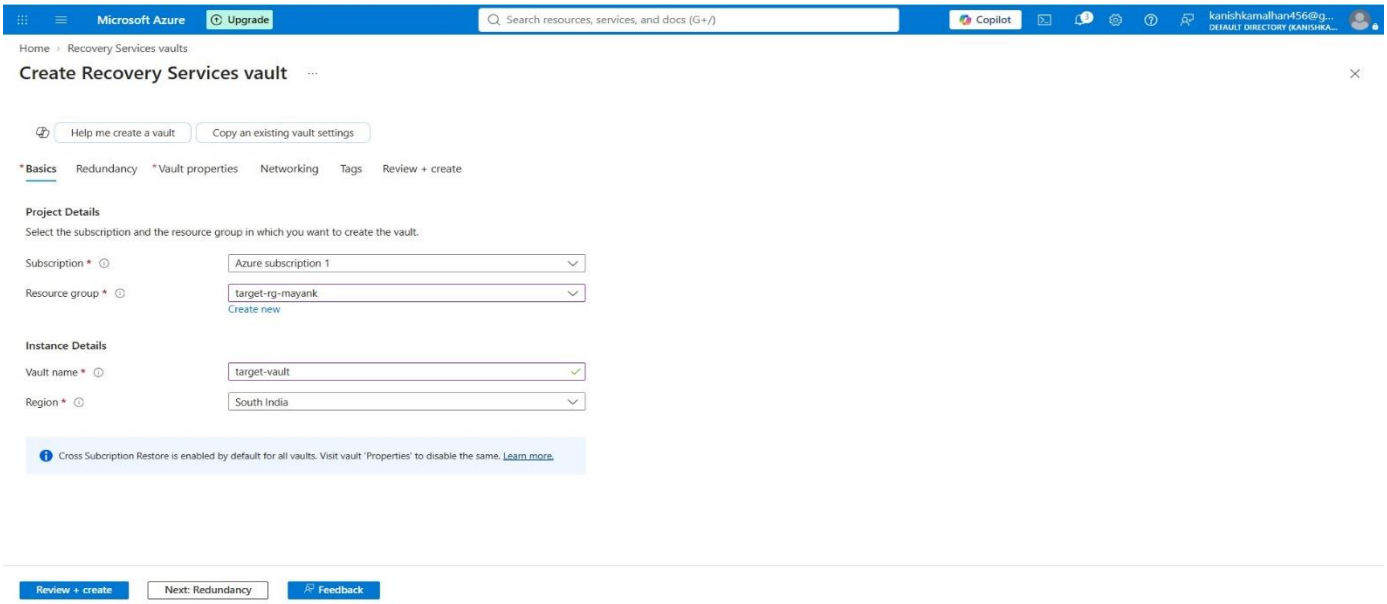
Step 2: Review the Target Resource Group

Open target-rg-mayank and verify that target-vnet is available in South India for the recovery environment.



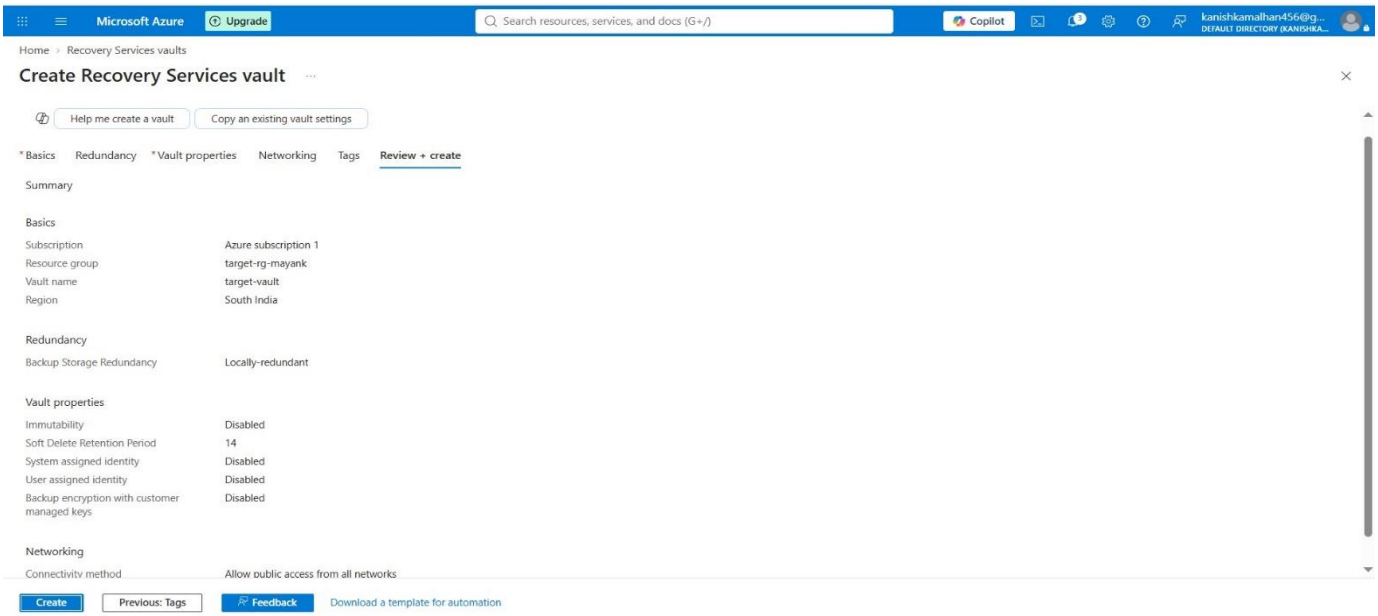
Step 3: Create the Recovery Services Vault

Start creating a Recovery Services vault. Select the subscription and target resource group, enter target-vault as the vault name, and select South India as the region.



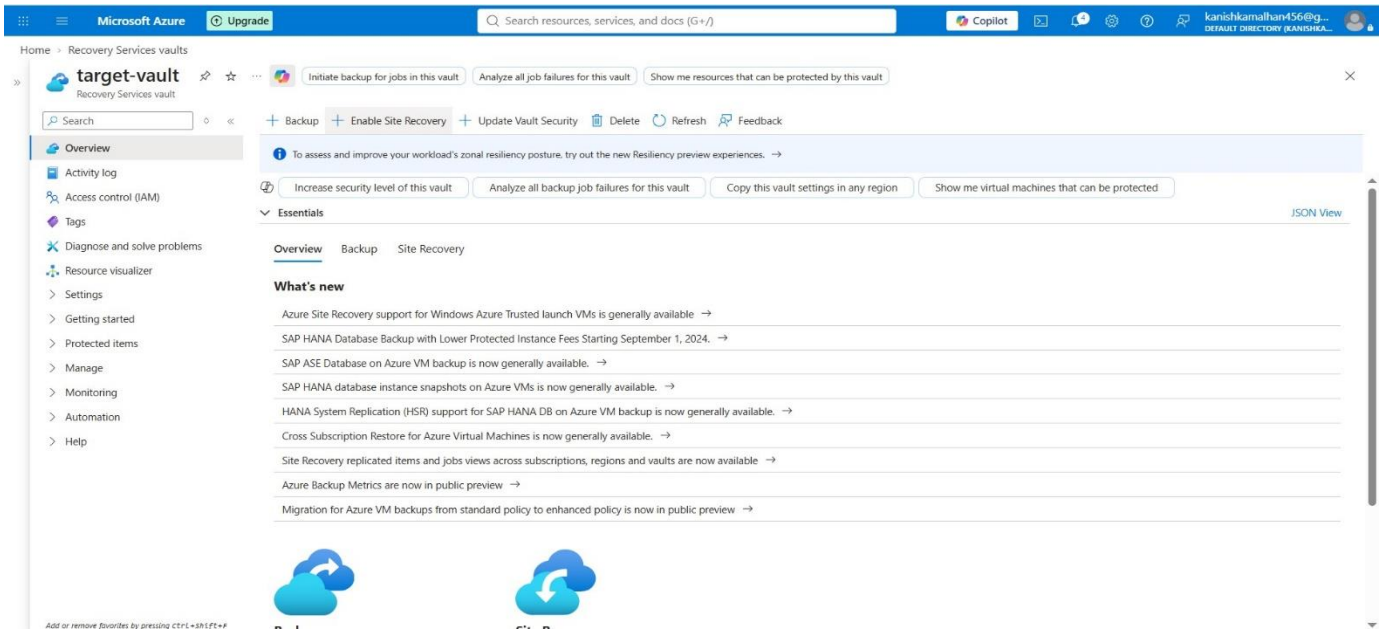
Step 4: Review and Create the Target Vault

Review the vault configuration, including the target resource group, South India region, and locally redundant storage (LRS). Select Create to deploy the vault.



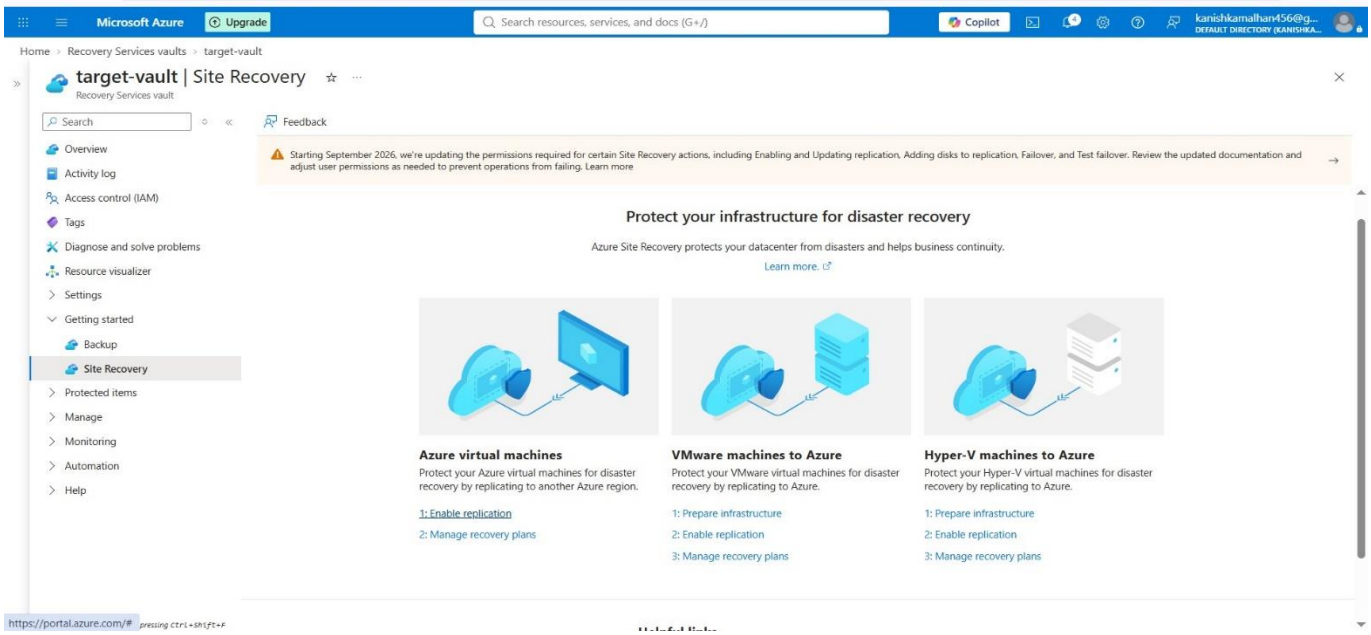
Step 5: Open the Target Recovery Services Vault

Open target-vault after deployment. This vault will be used to configure Azure Site Recovery for disaster recovery.



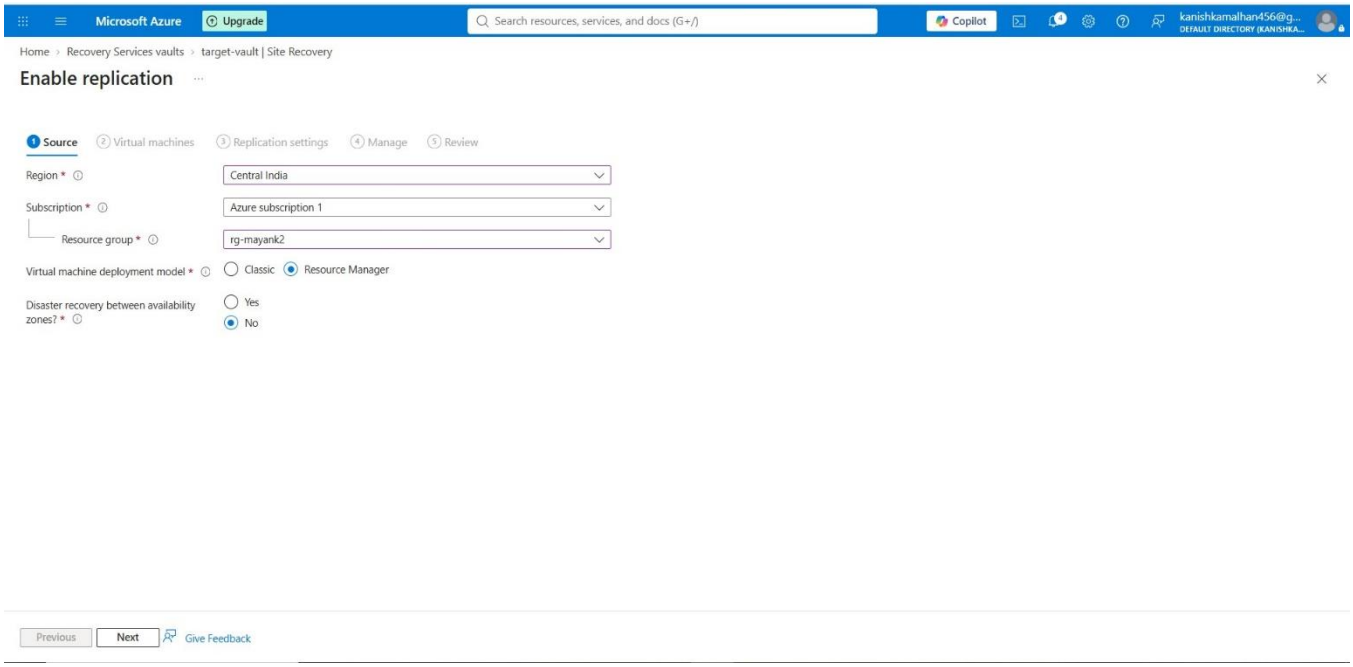
Step 6: Open Site Recovery

In target-vault, open Getting started > Site Recovery. Under Azure virtual machines, select Enable replication.



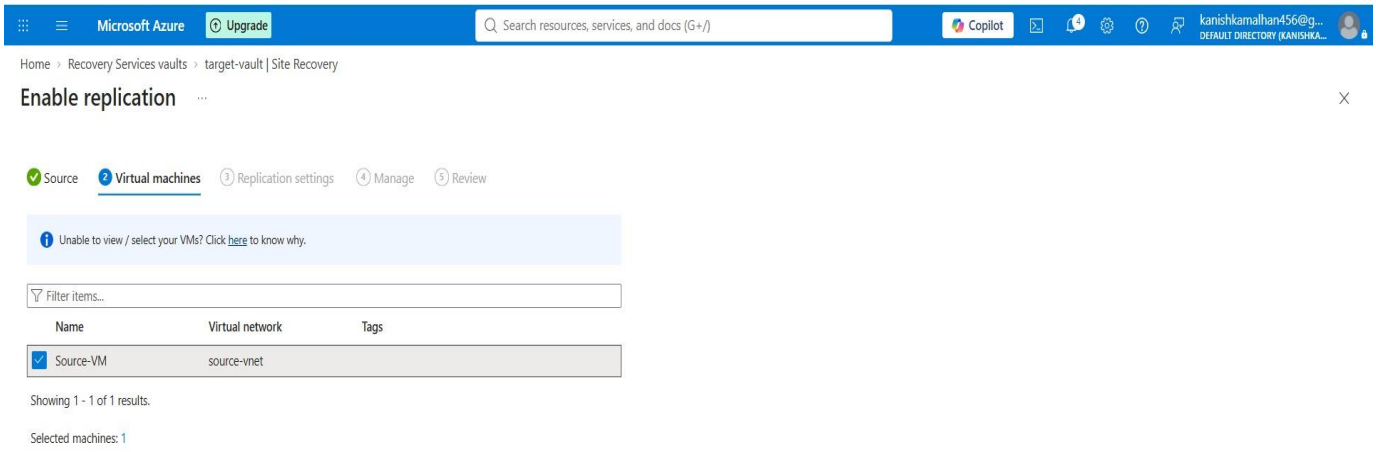
Step 7: Configure the Source for Replication

In the Enable replication wizard, select Central India as the source region, choose the subscription and rg-mayank2, select Resource Manager, and choose No for disaster recovery between availability zones.



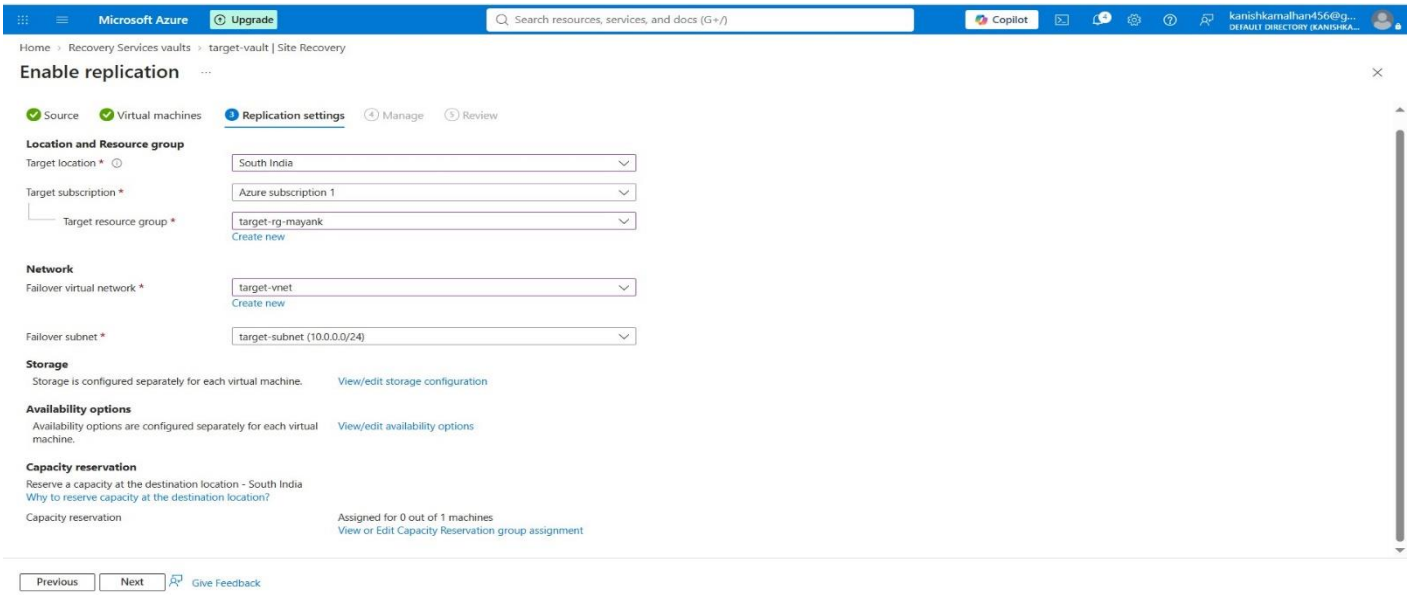
Step 8: Select the Source Virtual Machine

Select Source-VM from the list of virtual machines to replicate, then continue to the replication settings.



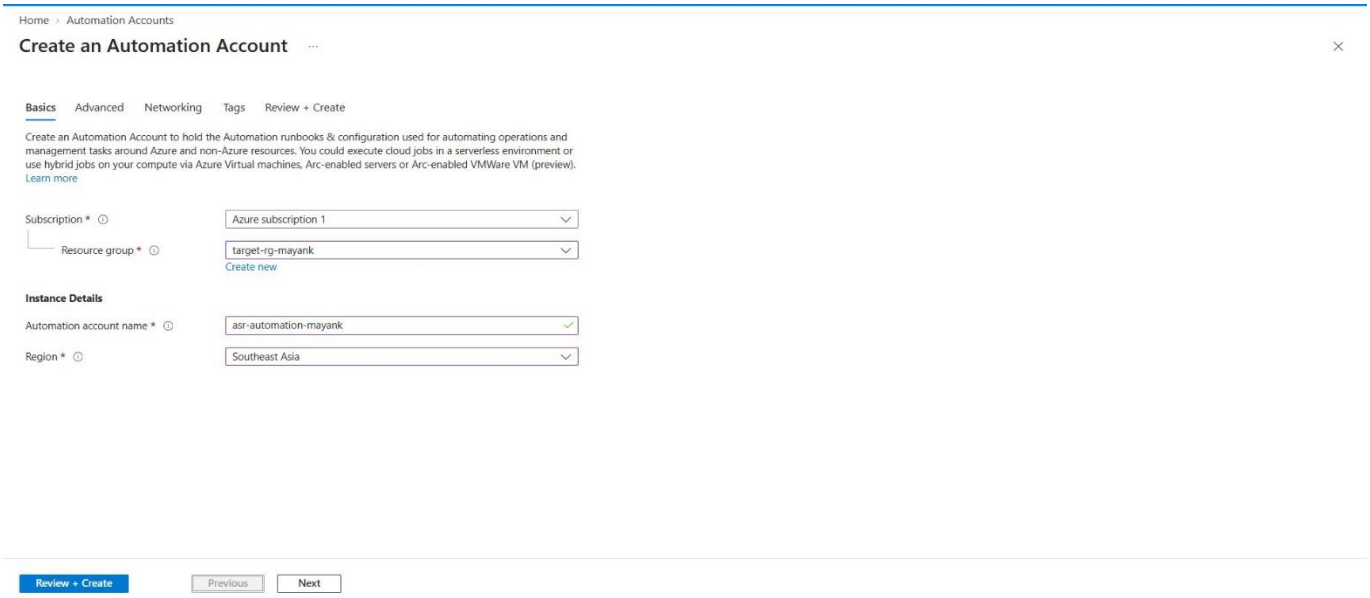
Step 9: Configure Replication Settings

Set South India as the target location. Select target-rg-mayank, target-vnet, and target-subnet as the failover destination resources.



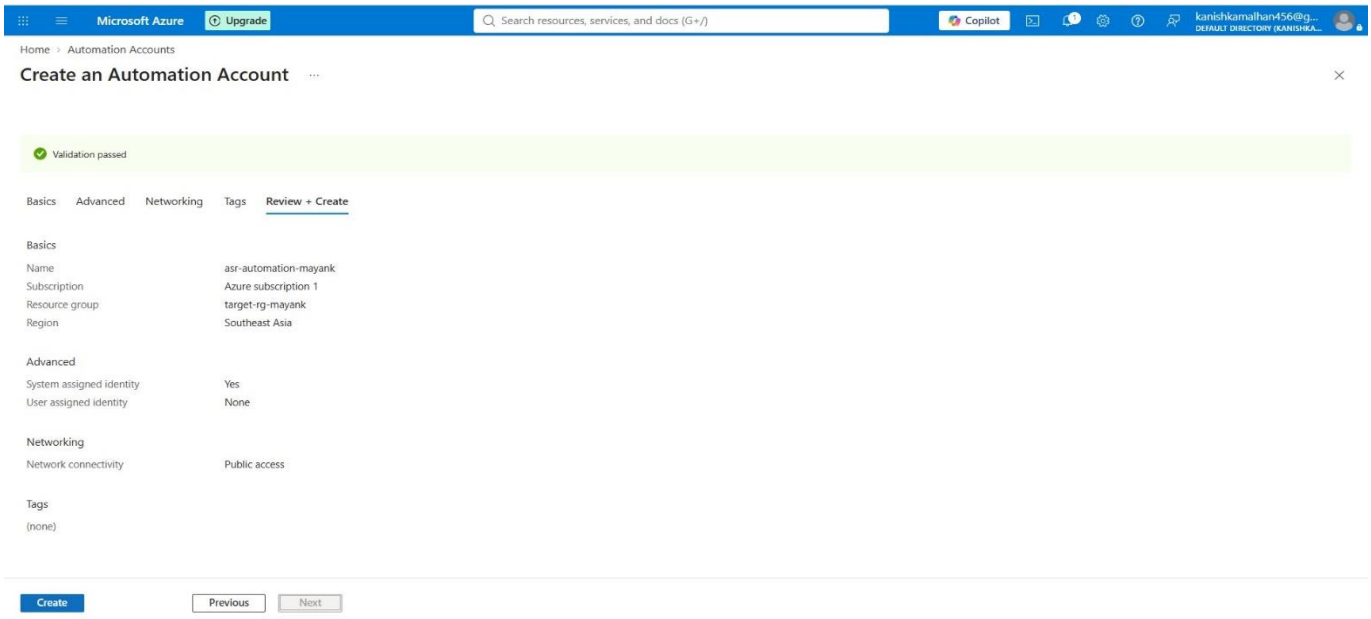
Step 10: Create the Automation Account

Create an Automation Account named asr-automation-mayank in target-rg-mayank. Select Southeast Asia as the region, as shown in the configuration.



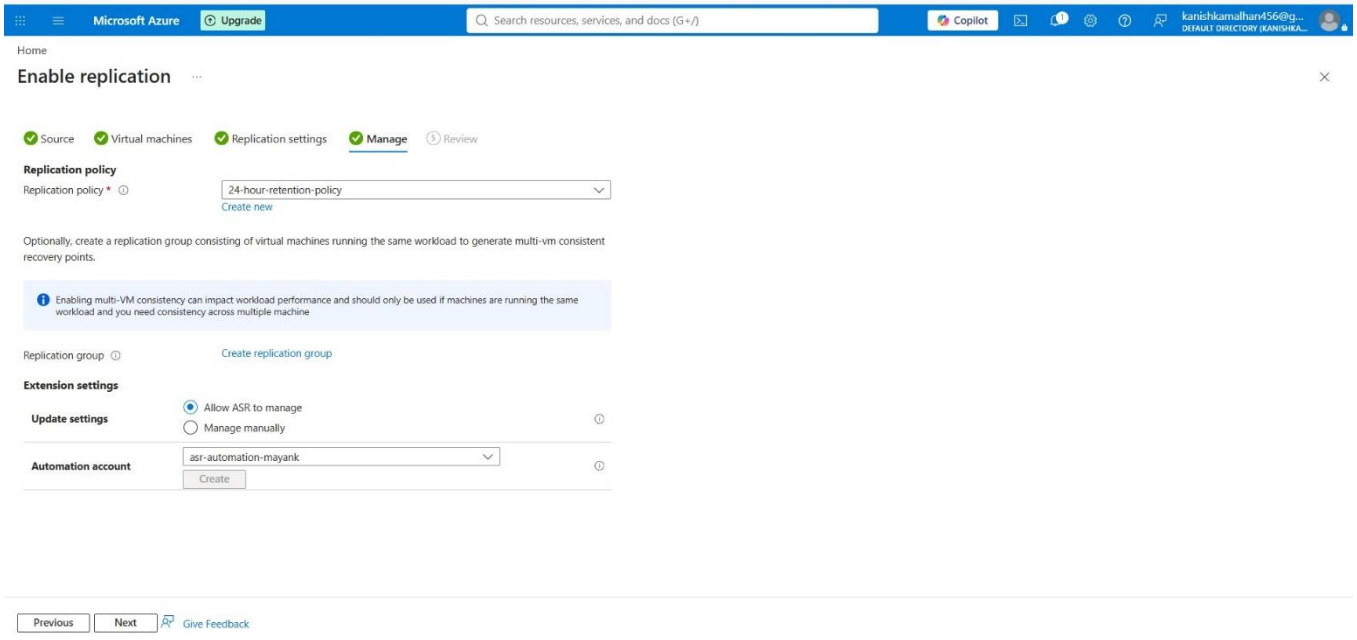
Step 11: Review and Create the Automation Account

Review the Automation Account settings, confirm validation has passed, and select Create.



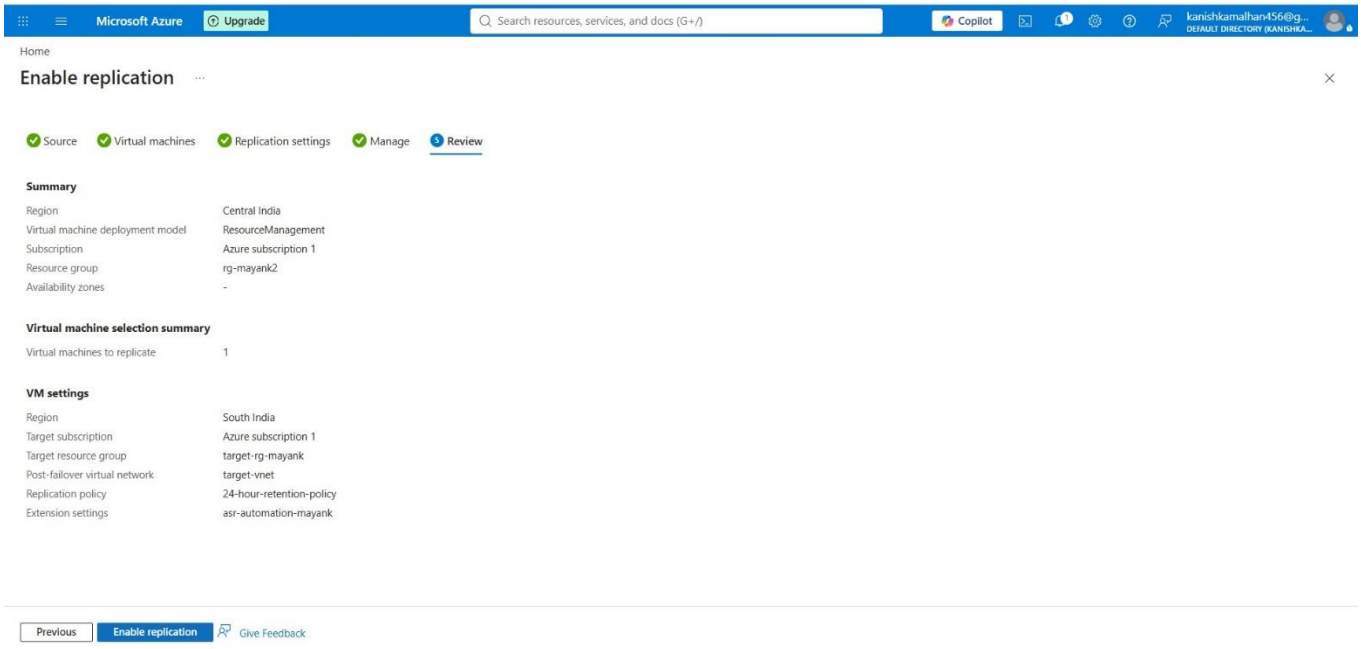
Step 12: Configure Replication Management Settings

On the Manage step, select the 24-hour-retention-policy replication policy. Allow Azure Site Recovery to manage extension updates and select asr-automation-mayank as the Automation Account.



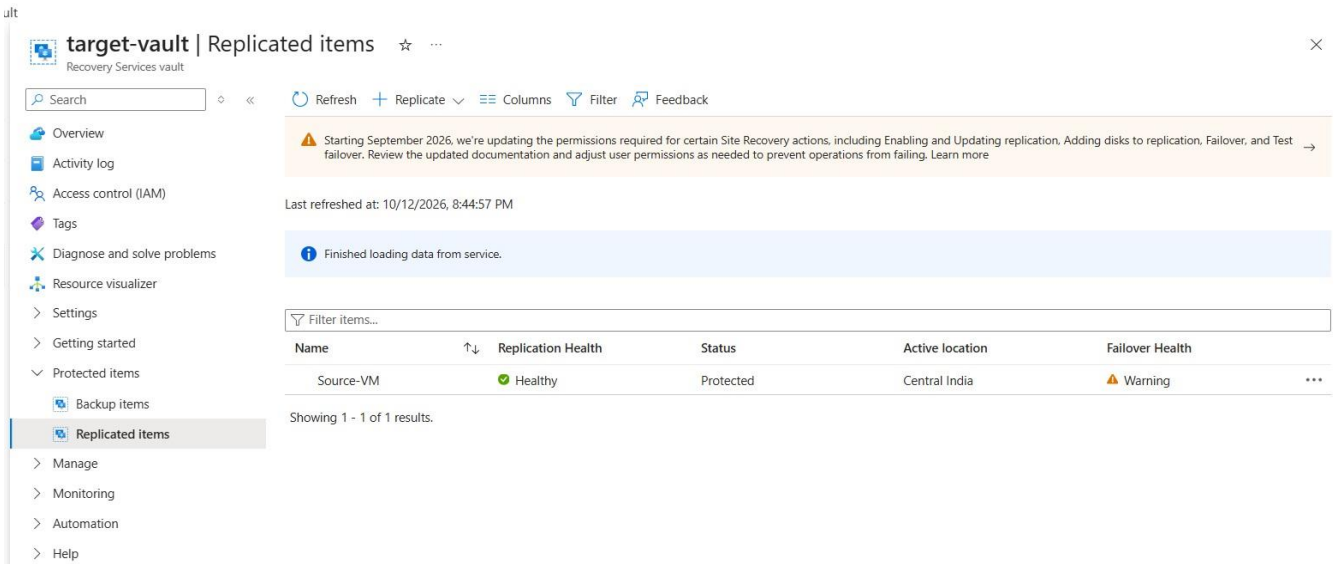
Step 13: Review the Replication Configuration

Review the replication summary. Confirm Source-VM is selected for replication, with South India, target-rg-mayank, target-vnet, the replication policy, and the Automation Account listed correctly. Select Enable replication.



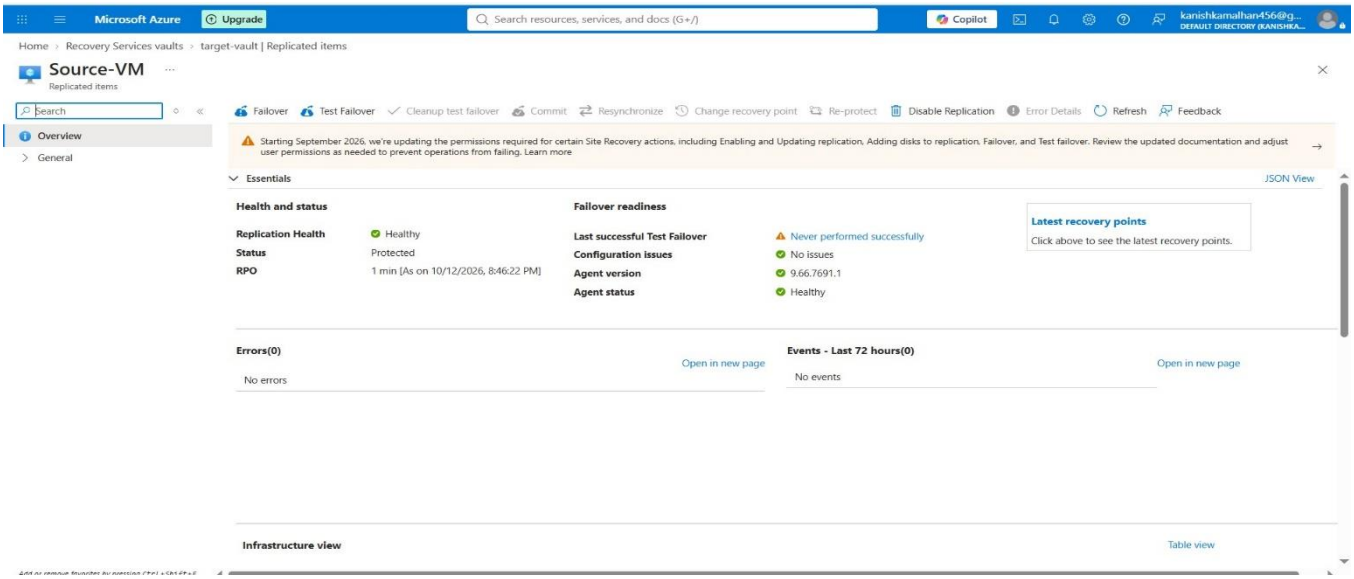
Step 14: Verify the Replicated Item

Open Protected items > Replicated items in target-vault. Verify that Source-VM appears in the list and check its replication health and status.



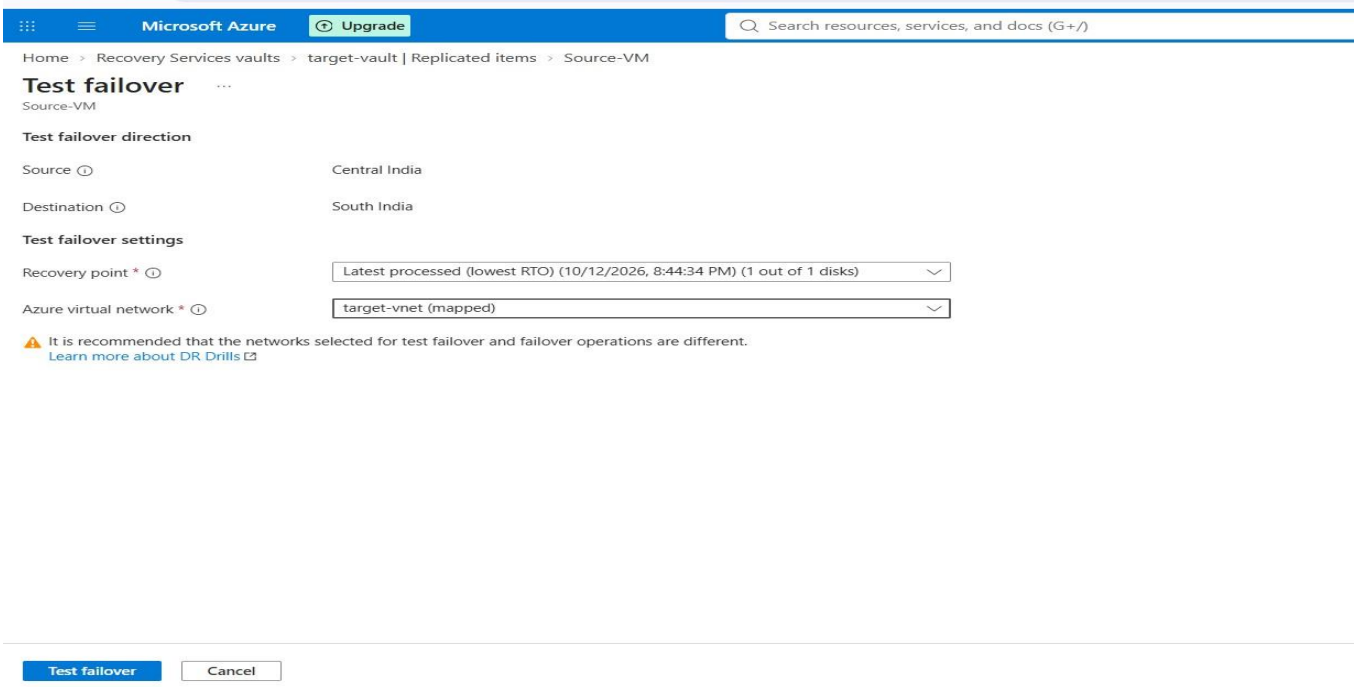
Step 15: Check Replication Health and Status

Open the Source-VM replicated item. Check that replication health is Healthy, status is Protected, the agent is healthy, and there are no configuration issues. The first screenshot may show that a test failover has not yet been performed.



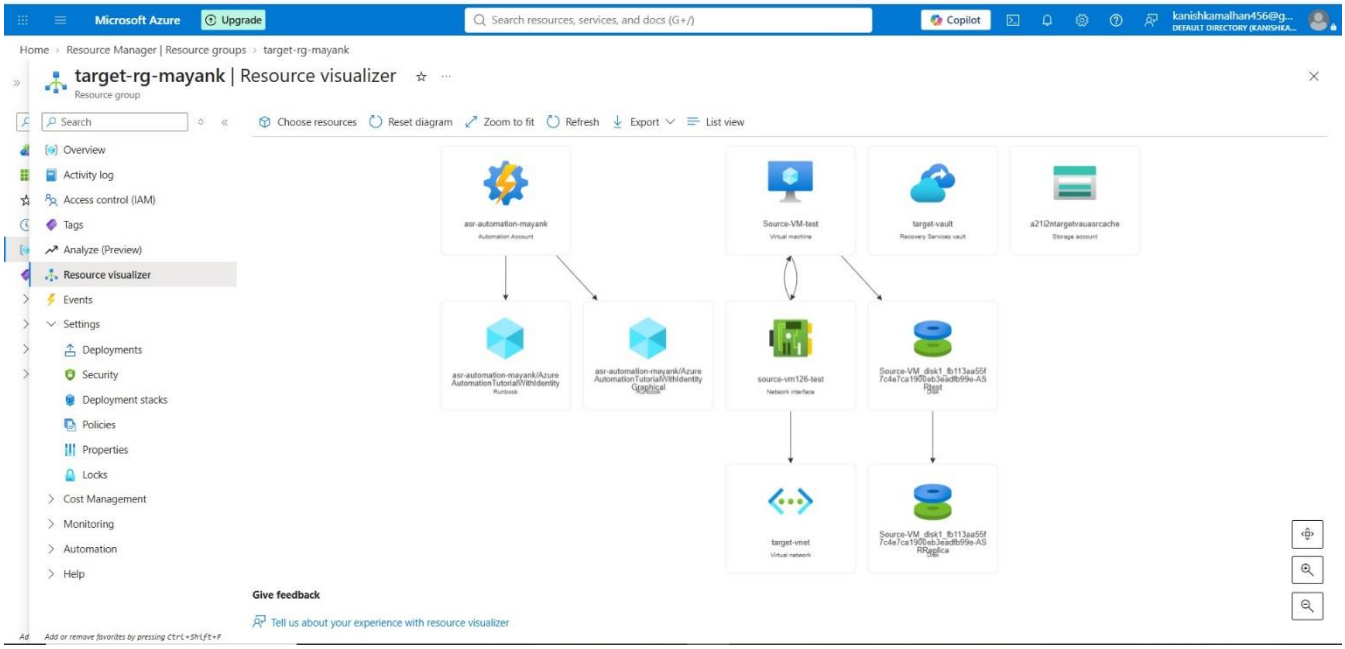
Step 16: Configure Test Failover

Select Test Failover. Confirm Central India is the source and South India is the destination, choose the latest processed recovery point, and select target-vnet for the test network. Start the test failover.



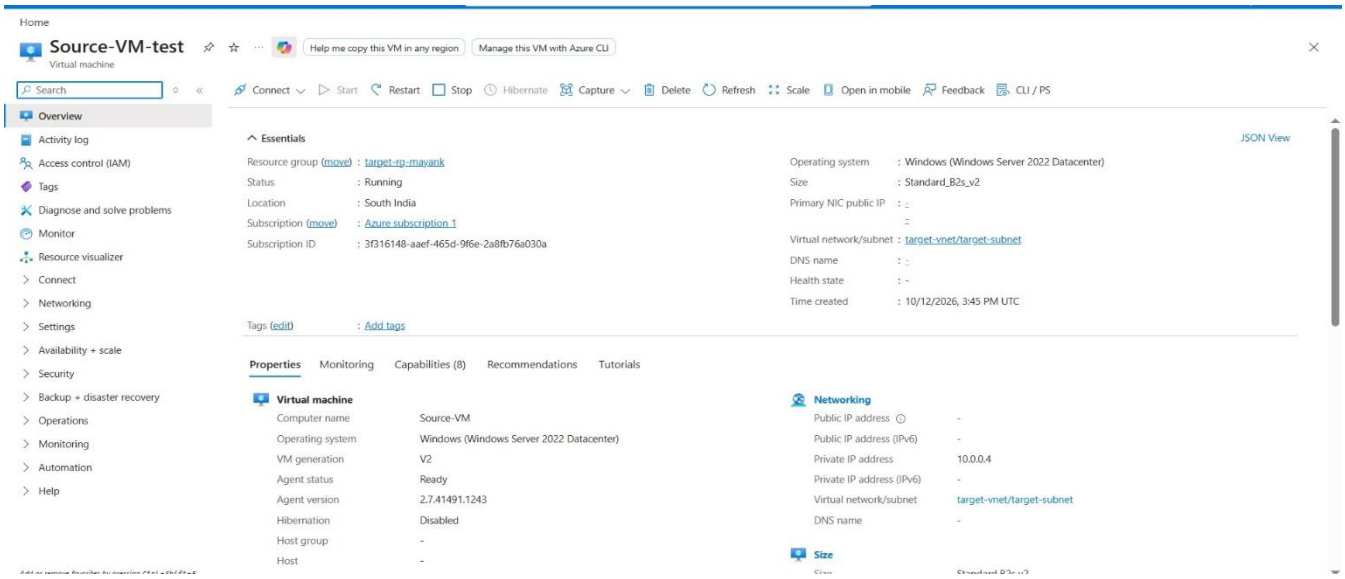
Step 17: Review the Test Failover Resources

Open the target resource group's Resource visualizer. Review the resources created for the test failover, including Source-VM-test, its network interface, target-vnet, and associated disks.



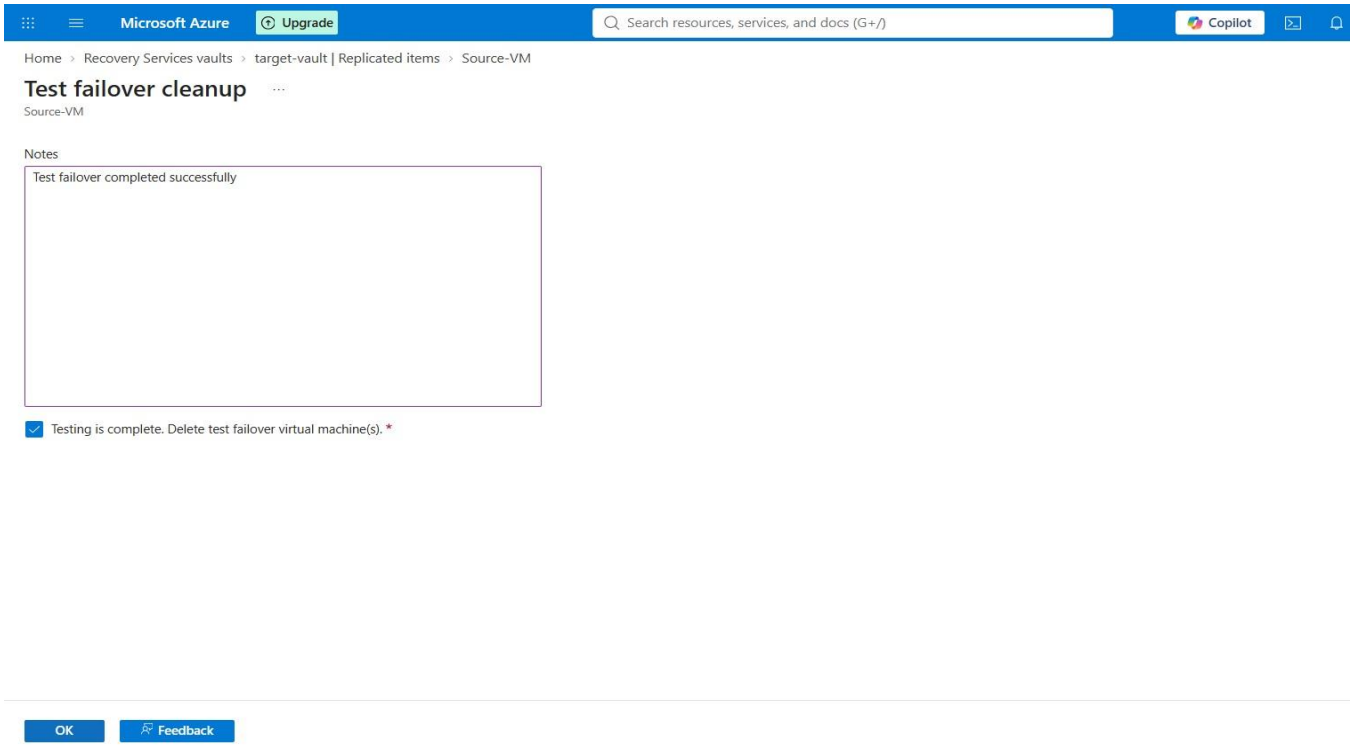
Step 18: Verify the Test Failover Virtual Machine

Open Source-VM-test and verify that the virtual machine is Running in South India and connected to target-vnet/target-subnet. This confirms that the test failover VM was created.



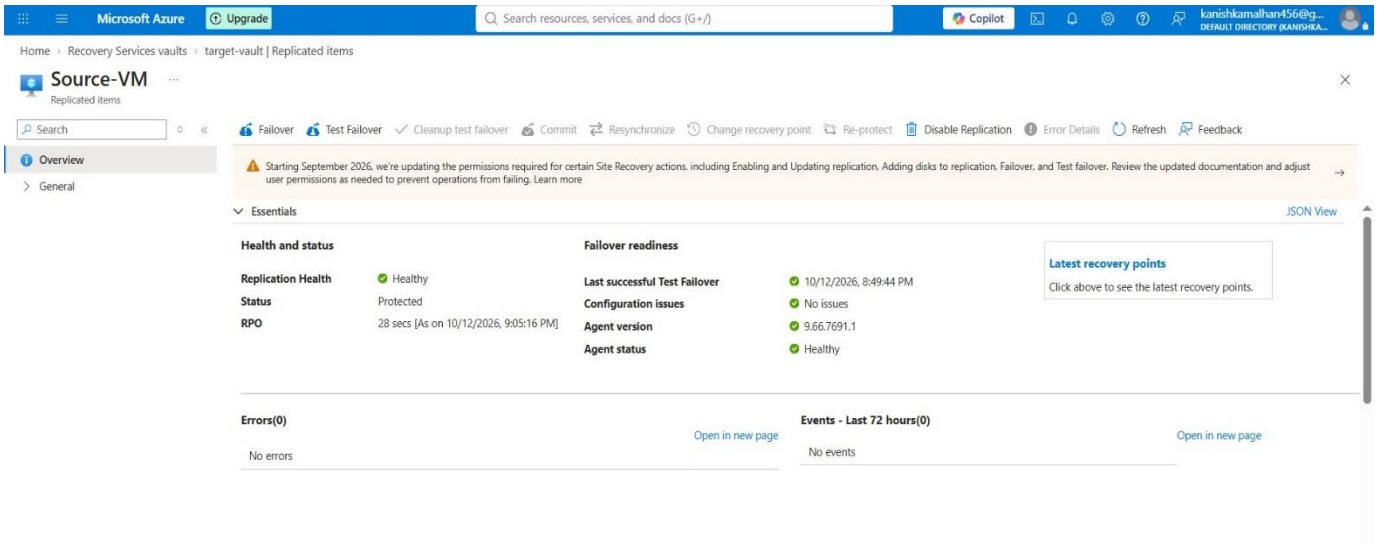
Step 19: Clean Up the Test Failover

After testing is complete, open Test failover cleanup. Enter a note such as “Test failover completed successfully,” keep the option to delete the test failover virtual machine selected, and select OK to clean up the test environment.



Step 20: Confirm Test Failover Success

Return to the Source-VM replicated item and verify that the last successful Test Failover time is displayed, replication health is Healthy, status is Protected, and configuration issues show No issues.



Verification And Result

- Source-VM is listed under Replicated items in target-vault.
- Replication health is Healthy and the VM status is Protected.
- The test failover VM, Source-VM-test, is created in South India and reaches Running status.
- The replicated item records a successful test failover and shows no configuration issues.
- The test failover VM is cleaned up after verification to avoid leaving unnecessary resources running.

Cleanup Note

The screenshots show cleanup of the test failover VM. If the lab is no longer required, disable replication and remove remaining lab resources only after confirming that all required evidence has been saved. Do not delete the source VM unless it is intentionally part of the cleanup.